

Performance evaluation and microstructural characterization of GFRP bars in seawater-contaminated concrete



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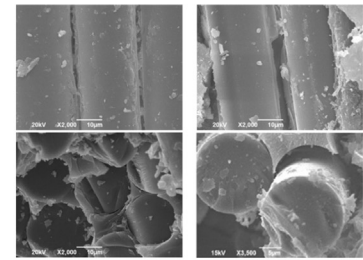
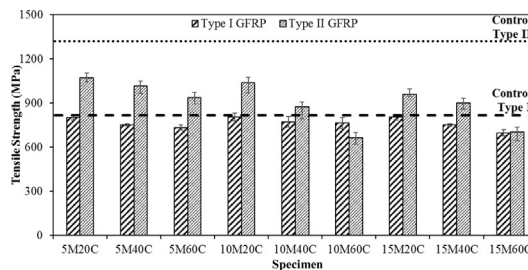
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HIGHLIGHTS

- Durability of GFRP bars in seawater-contaminated concrete was investigated.
- Strength retentions and changes in microstructural characteristics were recorded.
- Deterioration mechanisms of GFRP in seawater-contaminated concrete were reported.

GRAPHICAL ABSTRACT



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ABSTRACT

This paper aims to investigate the durability performance and microstructure characteristics of two different types of commercially-produced glass fiber-reinforced polymer (GFRP) bars conditioned in moist seawater-contaminated concrete. GFRP bars encased in seawater-contaminated concrete were immersed in tap water for 5, 10, and 15 months at temperatures of 20, 40, and 60 °C. Following conditioning, the GFRP bars were retrieved then tested to failure under uniaxial tension. Microstructure analysis was performed by employing differential scanning calorimetry (DSC), Fourier transform infrared (FTIR) spectroscopy, scanning electron microscopy (SEM), and matrix digestion using nitric acid. Type I GFRP bars, with the lower moisture uptake, exhibited insignificant strength reductions in the range of 2–15% whereas Type II, with the higher moisture uptake, exhibited substantial strength reductions in the range of 19–50%. Type II GFRP bars featured lower matrix retention and developed higher hydroxyl ions during conditioning than those of their Type I counterparts. The extent of degradation was more sensitive to the conditioning temperature rather than conditioning duration. A decrease in the glass transition temperature (T_g) of both types of GFRP bars was recorded, indicating matrix plasticization. Results of SEM highlighted matrix disintegration and fiber debonding after conditioning.

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1. Introduction

Glass fiber-reinforced polymer (GFRP) bars have a great potential to replace conventional steel reinforcement because of their light weight and corrosion resistance [1–3]. They are, however, susceptible to degradation in tensile properties when subjected

to alkaline or acidic solution, moisture/water, and elevated temperatures [1–3]. Although deterioration of GFRP in concrete occurs mainly due to alkali attack or resin plasticization by water uptake rather than chloride attack, saline solutions are considered a slightly more severe environment than fresh water [3]. Unless the long-term performance and degradation mechanism of GFRP bars are properly understood, it would be difficult to widely adopt such reinforcement by the construction industry.

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Much work has been carried out to study the durability performance of GFRP reinforcing bars in different environments using accelerate aging tests. Generally, simulated concrete-pore solutions have been employed to represent the actual service life in concrete [4–10]. GFRP bars were conditioned at temperatures ranging between 23 and 350 °C for durations up to 24 months. Test results indicated that higher temperatures and longer exposure led to significant loss in the tensile strength and reduction in the glass transition temperature (T_g). A decrease in the tensile strength was reported due to nonhomogeneous transfer of stress between adjacent fibers which was attributed to the delamination/debonding mechanism between fibers and polymer matrix [4]. It seemed, however, that the fiber, resin, and interface were unaffected by moisture and temperatures below 50 °C [4]. While irreversible chemical degradation of the polymer matrix was not detected, reversible plasticization was reported, leading to a lower T_g [4]. In another study, GFRP specimens were exposed to high temperatures up to 350 °C [5]. A decrease in the tensile strength was reported, associated with severe degradation of the polymer matrix and formation of microcracks. Conditioned GFRP bars experienced a 46 °C decrease in T_g [5]. Continuous immersion resulted in greater degradation in tensile properties than that caused by wet-dry cycles [8]. GFRP specimens preloaded up to 80% of the ultimate tensile strength prior to conditioning exhibited a decrease in T_g due to plasticization [11]. Moisture acted as a plasticizer, thus reducing the flexural strength and T_g of the matrix [12].

The in-situ behavior of GFRP reinforcing bars was examined in few studies [13–15]. To better resemble actual field conditions, GFRP specimens were wrapped with mortar and immersed in tap water. Compared with GFRP bars conditioned in simulated pore solution, the durability of mortar-wrapped counterparts was less affected by accelerated aging. Increasing the conditioning temperature had more detrimental effect on the tensile strength of GFRP bars rather than that caused by increasing the conditioning duration [13]. GFRP bars retrieved from concrete beams immersed in tap water for 3, 6, and 12, months lost 42, 56, and 65% of their strength, respectively [14]. Even in natural weathering, GFRP bars embedded in concrete beams for 18 and 30 months lost 35 and 40% of their tensile strength, respectively [14]. While cracked concrete beams with pre-tensioned GFRP strands retained nearly 15% of their ultimate strength after 9 months of exposure to wet/dry cycles, uncracked counterparts experienced a 30% strength retention after 18 months of similar exposure [15].

The combined effect of sustained loading and harsh environment on the performance of GFRP bars was investigated [4,16,17]. Debaiky et al. [4] evaluated the residual tensile properties of sand-coated GFRP bars in alkaline and water environments combined with sustained loading and elevated temperature. The test periods varied from one to four months under elevated temperature to hasten degradation and simulate extended service periods. The reduction in the tensile strength was in the range of 7–13% of the guaranteed strength. Based on the work of Tannous and Saadatmanesh [16] and Bakis et al. [18], it was concluded that the applied stress could intensify deterioration of GFRP bars embedded in concrete. Nkurunziza et al. [17] evaluated the creep behavior of GFRP bars surrounded by either an alkaline solution (pH 12.8) or de-ionized water (pH 7.0) at two levels of sustained tensile stresses of 25 and 38% of the guaranteed tensile strength. No creep rupture was observed in this study over 417 days of conditioning under the two levels of sustained stresses. Alkaline solution tended to have more harmful effects on GFRP bars than those caused by de-ionized water at higher stress levels. The residual tensile strength of the GFRP bars after extended exposure to de-ionized water was almost unchanged. The alkaline solution caused, however, the residual strength to drop to 88% and 68% of the guaranteed strength under 25 and 38% stress levels, respectively [17].

Benmokrane et al. [19] examined the durability of GFRP reinforcing bars exposed to various sustained tensile loading and different alkaline environments. The level of applied stress changed the deterioration mechanism of GFRP: (1) at low stress levels, the degradation was affected by the diffusion rate of alkaline solution; (2) at higher stress levels, it was controlled by solution transport through resin cracks [19]. Almusallam and Salloum [20] reported similar findings where stressed GFRP bars imbedded in concrete showed an additional 18% reduction in the tensile strength relative to that of their unstressed counterparts. Stressed GFRP bars embedded in concrete beams then exposed to severe environments experienced higher reduction in their tensile strength compared to that of similar bars left in natural conditions [21]. Trejo et al. [22] reported that GFRP bars imbedded in unsaturated concrete for 7 years exhibited significantly lower rates of tensile strength reduction compared with that of GFRP bars subjected to accelerated aging tests.

Although GFRP bars have a great potential to replace conventional steel and overcome corrosion problems in offshore structures and bridge decks in coastal cities or seaports, evaluation of their durability performance and microstructural characteristics when conditioned in moist seawater-contaminated concrete has received little attention. This paper aims at filling this gap and providing insight into the durability performance of two different types of commercially-produced GFRP bars in severe environment. Tensile strength retentions of conditioned GFRP bars were measured to evaluate their durability performance. The tensile strength results are supplemented by rigorous microstructural analysis. The interaction between the void content, moisture uptake, matrix retention, fiber-matrix debonding, tensile strength retention, and the increase in hydroxyl ions caused by hydrolysis is elucidated.

2. Materials and methods

2.1. GFRP bar specimens

Two types of GFRP bars made of continuous glass fibers impregnated in epoxy resin were utilized in this study. The two GFRP bar types had ribs on the surface as shown in Fig. 1. Type I GFRP bars had inner and outer diameters of 7.2 and 8 mm whereas those of Type II were 8 and 9 mm, respectively. The average cross-sectional area of each type was determined according to the test method specified by the ACI 440.3R-12 [2]. Type I GFRP bars had an average cross-sectional area of 45 mm² whereas Type II had an average area of 54 mm². The void contents of Type I and Type II GFRP bars were determined as 0.10 and 0.23%, respectively, as per ASTM D3171 and D2734 [23,24]. The mass fraction of glass fibers, determined by matrix digestion using nitric acid, in accordance with ASTM D3171 [23], was 78.3% for Type I and 75.5% for Type II. It is customary that the fiber content calculated as per ASTM D3171 [23] includes fibers and fillers. Thermogravimetry analysis (TGA) was also performed following ASTM E1868 [25] to verify fiber contents determined by matrix digestion using nitric acid. TGA resulted in respective fiber contents of 79 and 75%, by mass. The difference in fiber content determined by both methods was in the range of 0.5–0.7%, which fell within the limits given in Section 14 of the ASTM E1868 [25]. The as-received tensile strength and modulus were determined according to the test method specified by the ACI 440.3R-12 [2]. Type I and Type II GFRP bars had as-received tensile strengths of 816 and 1321 MPa, respectively. The tensile modulus of both types was 53 GPa. The GFRP test specimens were 1200 mm long with the middle third surrounded by 50 × 50 mm



Fig. 1. GFRP bars used in the current study.

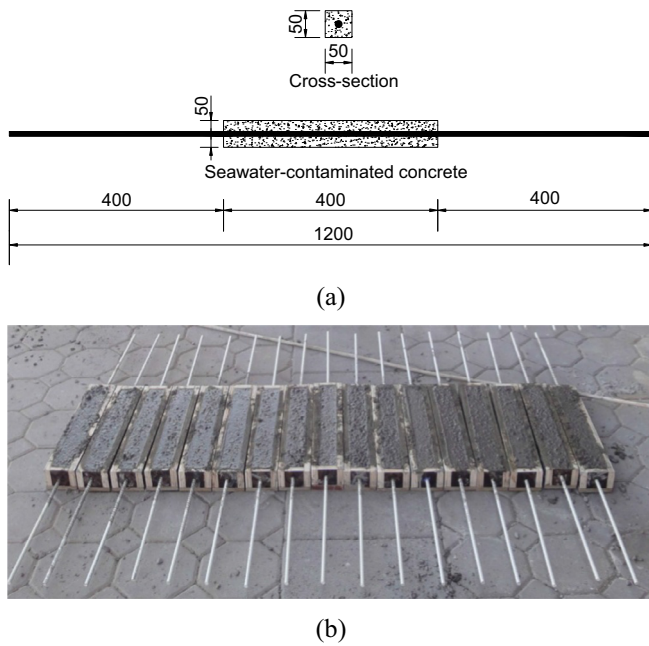


Fig. 2. Test specimens: (a) schematic, (b) after concrete casting.

seawater-contaminated concrete to represent concrete subjected to seawater splash in field condition (Fig. 2). Three replicate specimens were used for each testing condition.

2.2. Surrounding concrete

The mixture proportions of the concrete are shown in Table 1. Ordinary Portland cement (ASTM Type I) was used along with natural crushed stone coarse aggregate with a maximum size of 10 mm. The fine aggregate was a blend of dune sand and natural crushed stone mixed in 1:1, by mass. The mixing water was seawater obtained from the Arabian Gulf. Table 2 presents the physical and mechanical properties of surrounding concrete. The compressive and splitting tensile strengths, ultrasonic pulse velocity (UPV), and electrical resistivity were measured following ASTM C39, C496, and C597 and AASHTO TP95 [26–29]. The concrete displayed average compressive and tensile splitting strengths of 40 and 3.2 MPa, respectively. The corrosion protection category of the surrounding concrete was classified as ‘moderate’ given its electrical resistivity of 9.8 kΩ cm [30].

2.3. Conditioning tanks

The concrete-encased GFRP bar specimens were immersed in temperature-controlled tap water baths for 5, 10, and 15 months. Three conditioning tanks with built-in heaters and thermostat were fabricated to maintain constant water temperatures of 20, 40, and 60 °C during conditioning. Prior to conditioning, polyvinyl chloride (PVC) pipes were installed along the exposed parts of the GFRP bars then blocked/sealed carefully at each end to protect these regions during conditioning. The temperature of the water inside the tanks was monitored regularly during conditioning and any deviation in the temperature of the hot water was adjusted. Test

Table 1
Surrounding concrete mix proportions.

Unit weight (kg/m ³)				
Cement	Fine aggregate	Coarse aggregate	Seawater	w/c
400	580	1160	200	0.5

Table 2
Physical and mechanical characteristics of surrounding concrete.

Compressive strength (MPa)	Splitting strength (MPa)	Electrical resistivity (kΩ.cm)	UPV (m/s)
40	3.2	9.8	5405

specimens under accelerated aging are shown in Fig. 3. Results of conditioned specimens are compared with those of control unconditioned specimens kept at ambient conditions prior to testing.

2.4. Moisture absorption

The moisture uptake due to conditioning was measured in accordance with ASTM D570 [31]. Naturally, the change in mass before and after immersion represents the water absorbed during conditioning. However, the GFRP reinforcing bars may have experienced mass dissolution due to degradation mechanisms including hydrolysis reaction. In order to account for this mass loss, conditioned specimens were dried in an oven at 100 °C for 24 h. The recorded oven-dried mass was compared to the initial mass of GFRP samples. Eq. (1) was used to determine the corrected moisture uptake by mass:

$$\text{Moisture uptake (\%, by mass)} = 100 \times \frac{(\text{Conditioned mass} - \text{Oven dried mass})}{(\text{Initial mass})} \quad (1)$$

2.5. Tensile strength

Following conditioning, the GFRP bars were retrieved carefully from the concrete prisms. The PVC pipes were removed and end grips were installed. The end grips consisted of a steel pipe, 400 mm long, with inner and outer diameters of 25 and 34 mm, respectively as shown in Fig. 4(a). The steel pipe was cleaned carefully and randomly threaded from inside to make the inner surface rough. The pipe was then installed at each end of the specimen then filled with epoxy to maintain adequate bond between the GFRP bar and the inner surface of the pipe. The test region of the GFRP bar, middle 400 mm, was instrumented with three strain gauges to monitor the strains during tensile testing. One strain gauge was installed at the midpoint of the test region. The other strain gauges were installed at a distance 100 mm away from each end of the test region. The specimens were tested to failure under uniaxial tension in displacement control at a rate of 1.5 mm/min as shown in Fig. 4(b). A data acquisition system was used capture the load and strain readings.

2.6. Microstructure evaluation

Microstructure and morphological changes of control and conditioned GFRP samples were examined using a JEOL-JSM 6390A scanning electron microscope (SEM). Samples were taken from the middle 100 mm of the specimens, polished, and sputter coated with a thin gold layer (99.9% purity). The layer was thin enough (2–5 nm) to provide conductivity during testing without concealing the surface morphology. The obtained micrographs were used to identify possible degradation in fibers, polymer, and fiber-matrix interface.

To evaluate the fiber and matrix contents of conditioned GFRP reinforcing bars, matrix digestion using nitric acid was utilized, according to ASTM D3171 [23]. While the chemical reaction of GFRP bars with nitric acid results in digestion and dissolution of the epoxy resin, the fibers remain unaffected. Samples of 0.5–1.5 g were collected and mixed with 50 mL of 70% aqueous nitric acid. The mixture was then placed in a controlled temperature bath for 6 h at 80 °C. The contents were filtered and washed with distilled water prior to drying in an oven for 1 h at 100 °C. Eq. (2) gives the matrix content in terms of mass measured.

$$\text{Matrix content (\%, by mass)} = 100 \times \frac{(\text{Initial mass} - \text{Final mass})}{(\text{Initial mass})} \quad (2)$$

Fourier transform infrared (FTIR) spectroscopy was performed on GFRP samples taken from the middle 100 mm of the specimens to measure the degradation due to hydrolysis reaction. Powdered specimens were interground with potassium bromide (KBr) at a powder sample:KBr = 1:4, by mass. The mixed powder was analyzed from 400 to 4000 cm^{−1} at a resolution of 1 cm^{−1} using a Varian 3100 FT-IR spectrometer.

Powdered GFRP samples taken from the middle 100 mm of the specimens were examined by differential scanning calorimetry (DSC) using a DSC Q2000 calorimeter equipped with a refrigerated cooling system. Specimens, on the range of 10 mg, were sealed in aluminum pans and prepared for analysis. DSC curves were obtained by heating the samples from 20 to 225 °C at a heating rate of 5 °C/min. The glass transition temperature (T_g) was determined according to ASTM E1356 standard [32]. Two scans were performed for each specimen. The first scan was used to compare the T_g of conditioned samples to that of the unconditioned control ones. A decrease in T_g after the first scan would indicate matrix plasticization. The second scan is useful to identify the mechanism of degradation. If the T_g of a conditioned GFRP sample, after the second scan, is in the same range as that of the control sample, it would indicate a reversible plasticizing effect due to moisture absorption. However, a lower T_g compared to that of the control would signify an irreversible chemical degradation.

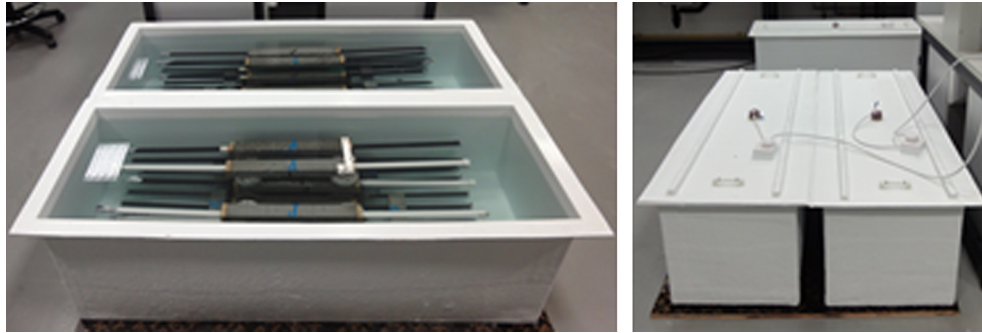


Fig. 3. Test specimens under accelerated aging.

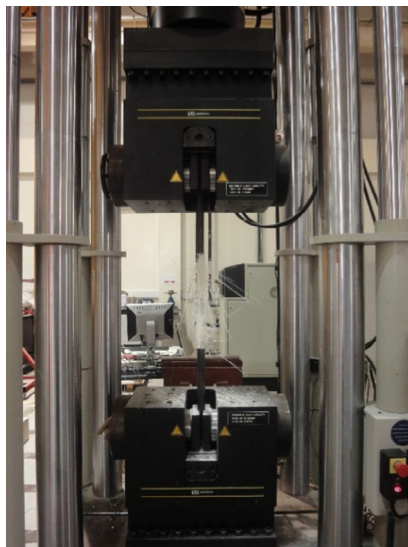
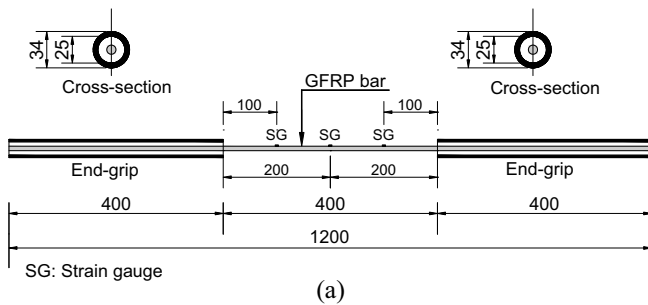


Fig. 4. Test set-up and instrumentation: (a) end grips and strain gauges, (b) tensile test set-up.

3. Experimental results and discussion

3.1. Moisture uptake

Generally, the moisture could penetrate into a composite material by the flow of water into microgaps between polymer chains, capillary transport into microcracks at the fiber-matrix interface, and through matrix microcracks that formed during the compounding process [33]. Fig. 5 shows the moisture absorption of Type I and II GFRP bars due to conditioning. For both types of GFRP bars, conditioning at 20 °C for 5 months did not result in any moisture uptake. GFRP bars conditioned at 20 °C exhibited, however, some moisture uptake at 10 and 15 months of exposure. GFRP bars

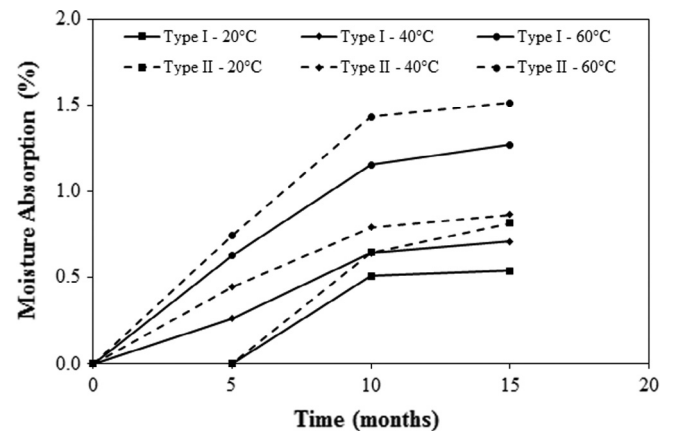


Fig. 5. Moisture uptake of GFRP bars.

conditioned at the higher temperatures of 40 and 60 °C exhibited a moisture uptake from the beginning of the accelerated aging test. They had higher moisture uptake than that of their counterparts conditioned at 20 °C at all times of exposure.

From Fig. 5, it can be seen that increasing the conditioning temperature resulted in more water absorption due to a higher diffusion rate and formation of microcracks. The moisture uptake increased rapidly within the first 10 months of exposure, particularly for the specimens conditioned at the higher temperatures of 40 and 60 °C. Further increase in the time of conditioning resulted in no or insignificant additional increase in the moisture uptake. This suggested that moisture absorption had possibly reached an equilibrium state after 10 months of conditioning.

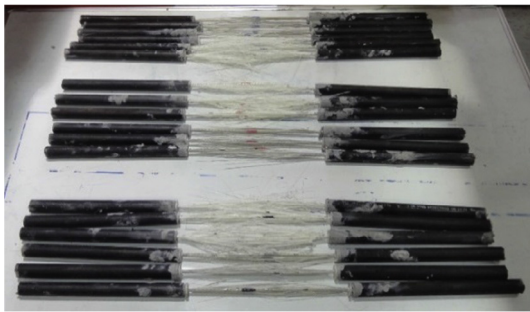
For any conditioning regime, Type II GFRP bars experienced higher moisture uptake than that exhibited by their Type I counterparts. Type II GFRP bars had higher void content than that of Type I, which resulted in higher moisture absorption during conditioning. The increased moisture uptake exhibited by Type II GFRP bars facilitated progression of the hydrolysis reaction (see FTIR results). Hydrolysis causes matrix softening and impairs the bond at the fiber-matrix interface.

3.2. Tensile strength

All specimens failed by violent rupture of fibers accompanied by fiber debonding at the fiber-matrix interface (see Fig. 4(b)). None of the tested GFRP bars exhibited premature failure at the end grips. Photos of unconditioned and conditioned GFRP bars after the tensile test to failure are given in Fig. 6(a) and (b), respectively. These photos confirm failure of the GFRP bars within the test region. The GFRP bars of both types exhibited an almost linear stress-strain



(a)

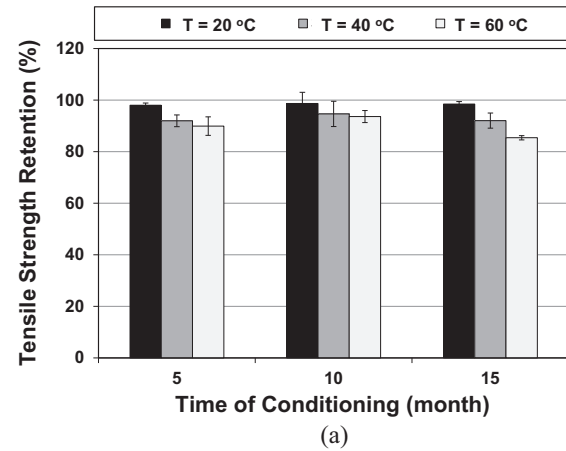


(b)

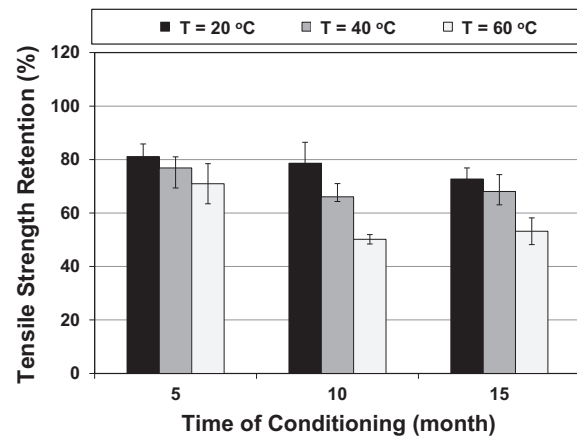
Fig. 6. Photos of tested GFRP bars: (a) unconditioned bars, (b) conditioned bars.

response up to failure. The tensile strengths of conditioned specimens are compared to those of the control unconditioned counterparts in Fig. 7. The tensile strength was calculated by dividing the maximum load by the average cross-sectional area of the GFRP bar determined according to the test method specified by the ACI 440.3R-12 [22]. The specimen designation shown in this figure consists of four characters. The first two characters refer to the conditioning duration (5 M, 10 M, and 15 M) whereas the last two characters refer to the conditioning temperature (20C, 40C, and 60C).

Tensile strength retentions of Type I GFRP bars are shown in Fig. 8(a). Type I GFRP exhibited insignificant tensile strength reductions of 2, 8, and 10% after 5 months of conditioning at 20, 40, and 60 °C, respectively. Subsequently, the tensile strength retention of



(a)



(b)

Fig. 8. Tensile strength retention of GFRP bars: (a) Type I, (b) Type II.

Type I bars remained almost unaltered till the end of the accelerated aging test, except for the specimens conditioned at 60 °C for 15 months. These specimens experienced a tensile strength reduction of 15%. The tensile strength retention of Type I GFRP bars tended to decrease with an increase in the temperature.

Tensile strength retentions of Type II GFRP bars are shown in Fig. 8(b). Type II GFRP bars experienced inferior durability performance in moist seawater-contaminated concrete than that

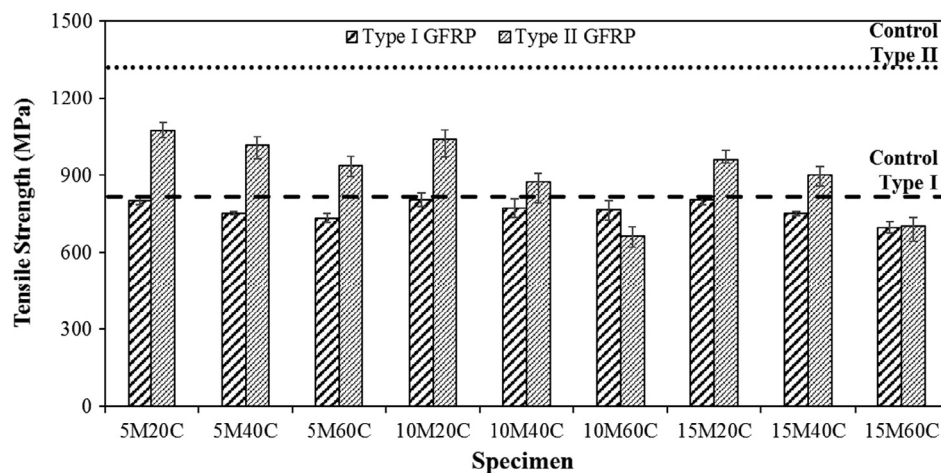


Fig. 7. Tensile strength of conditioned GFRP bars.

exhibited by Type I. At 5 months of accelerated aging, Type II bars experienced tensile strength reductions of 19, 23, and 29% at 20, 40, and 60 °C, respectively. The tensile strength retention decreased as the time of conditioning increased from 5 to 10 months at all temperatures. At 10 months of conditioning, Type II bars experienced tensile strength reductions of 21, 34, and 50% at 20, 40, and 60 °C, respectively. The reduction in the tensile strength was more pronounced at the higher temperatures. This demonstrates that hydrolysis is accentuated at high temperature, which would result in the disintegration of chemical bonds at the fiber-matrix interface and separation between the fiber and matrix. Increasing the conditioning time to 15 months further decreased the tensile strength retention of Type II GFRP bars conditioned at 20 °C where a strength reduction of 27% was recorded. The tensile strength retentions of Type II bars subjected to 15 months of conditioning at 40 and 60 °C were almost equal to those of their counterparts subjected to 10 months of conditioning. It seemed that the chemical bonds at the fiber-matrix interface was severely weakened after 10 months of conditioning at high temperatures by the hydrolysis reaction to the extent that an additional time of conditioning did not further reduce the strength retention.

Although Type II GFRP bars showed higher tensile strength prior to conditioning than that of Type I, the durability performance of the former was inferior compared with that of the latter. The tensile strength reduction could be due to matrix softening and/or fiber-matrix interfacial debonding, both caused by hydrolysis. Type II GFRP bars had higher void content and moisture uptake than those of Type I. The increased moisture absorption exhibited by Type II GFRP bars facilitated progression of the hydrolysis reaction (see FTIR results) and impaired the bond at the fiber-matrix interface (see SEM results). This could explain why Type II GFRP bars exhibited inferior durability performance than that of Type I. It should be also noted that the difference in the diameter of both GFRP bar types used in the present study is approximately 10%. This minor difference in bar diameter is too small to have an effect on the rate of degradation.

3.3. SEM analysis

SEM tests were performed to examine the effect of conditioning temperature on the morphology of GFRP bars conditioned for 15 months. Longitudinal and cross-sectional micrographs of Type I GFRP bars are shown in Figs. 9 and 10, respectively. Fig. 9(a) highlights adhesion of the matrix to the fibers with good bonding at fiber-matrix interface in the unconditioned control sample. After 15 months of conditioning at 20 °C, separation at the interface was detected as shown in Fig. 9(b). The associated gap width reached approximately 2 µm. This degradation could have resulted from crack propagation from the matrix to the interface as opposed to directly into the fiber [34]. Specimens conditioned at 40 °C exhibited a wider gap at the interface on the order of 5 µm as shown in Fig. 9(c). Fig. 9(d) indicates that although increasing the temperature from 40 to 60 °C did not widen the gap at the interface, it produced a relatively smoother fiber surface. This was an indication of matrix disintegration due to hydrolysis, which was accelerated in the hot water environment. Matrix disintegration would lead to a non-uniform distribution of load among fibers [4,35]. Cross-sectional micrographs of Fig. 10 show signs of interfacial debonding in conditioned specimens, particularly at the higher temperatures of 40 and 60 °C.

Longitudinal and cross-sectional micrographs of Type II GFRP bars are shown in Figs. 11 and 12, respectively. Fig. 11(a) of the unconditioned control sample identifies a rough surface of fibers with adequate bond to the matrix. Specimens conditioned at 20 °C for 15 months (Fig. 11(b)) experienced a small gap at the fiber-matrix interface. The increase in temperature to 40 °C increased the gap opening as shown in Fig. 11(c). Yet, it should be pointed out that the glass fiber remained unaffected by conditioning. The micrograph of the samples conditioned at 60 °C is presented in Fig. 11(d). Substantial disintegration of the matrix is noticed at the fiber-matrix interface with further development in the gap to a width of 6 µm. The cross-sectional micrographs shown in Fig. 12 show signs of interfacial debonding at the fiber-matrix

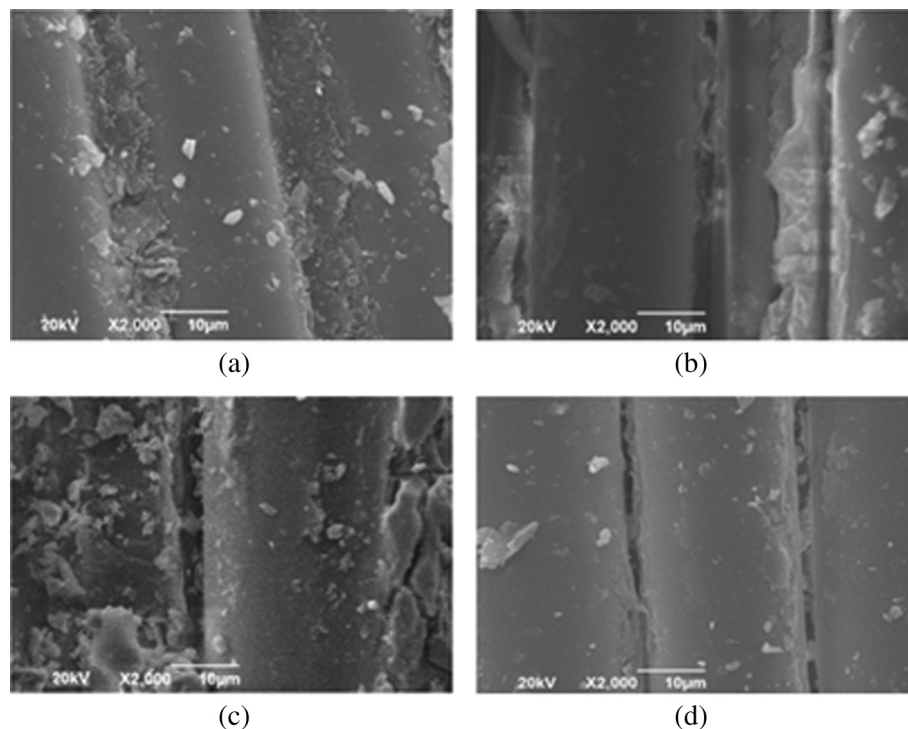


Fig. 9. Longitudinal micrograph of Type I GFRP bars: (a) control, (b) conditioned for 15 months at 20 °C, (c) conditioned for 15 months at 40 °C, (d) conditioned for 15 months at 60 °C.

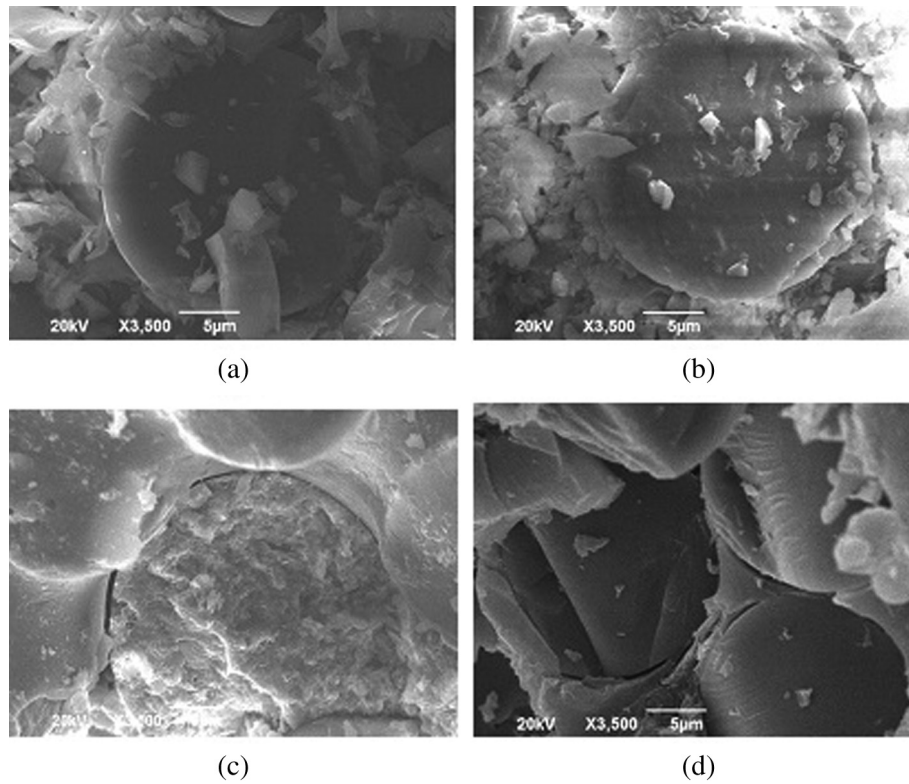


Fig. 10. Cross-sectional micrograph of Type I GFRP bars: (a) control, (b) conditioned for 15 months at 20 °C, (c) conditioned for 15 months at 40 °C, (d) conditioned for 15 months at 60 °C.

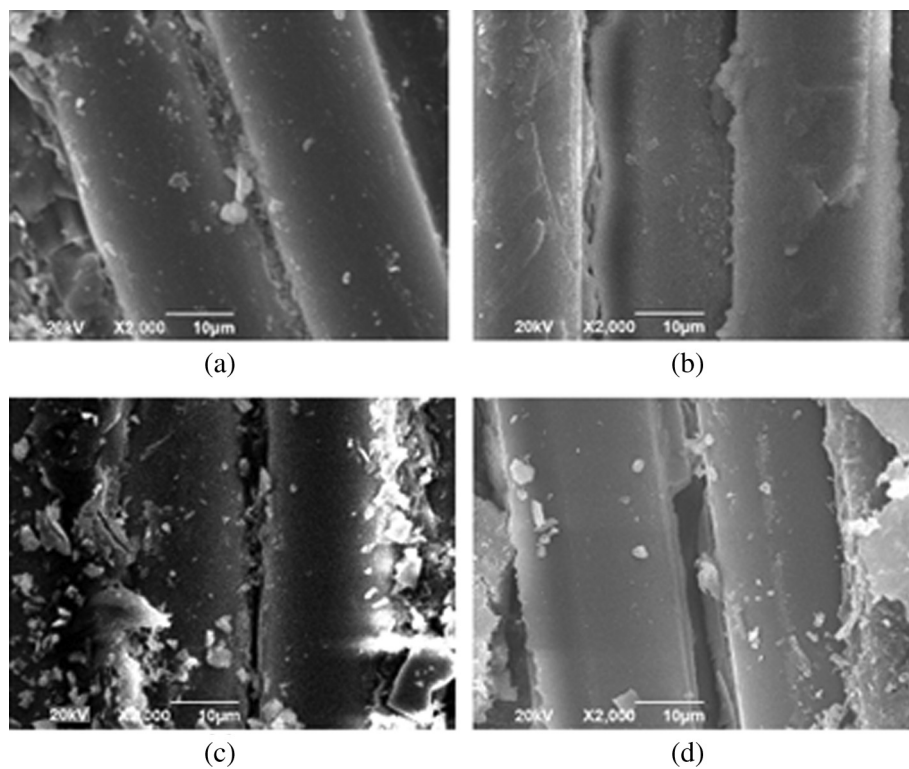


Fig. 11. Longitudinal micrograph of Type II GFRP bars: (a) control, (b) conditioned for 15 months at 20 °C, (c) conditioned for 15 months at 40 °C, (d) conditioned for 15 months at 60 °C.

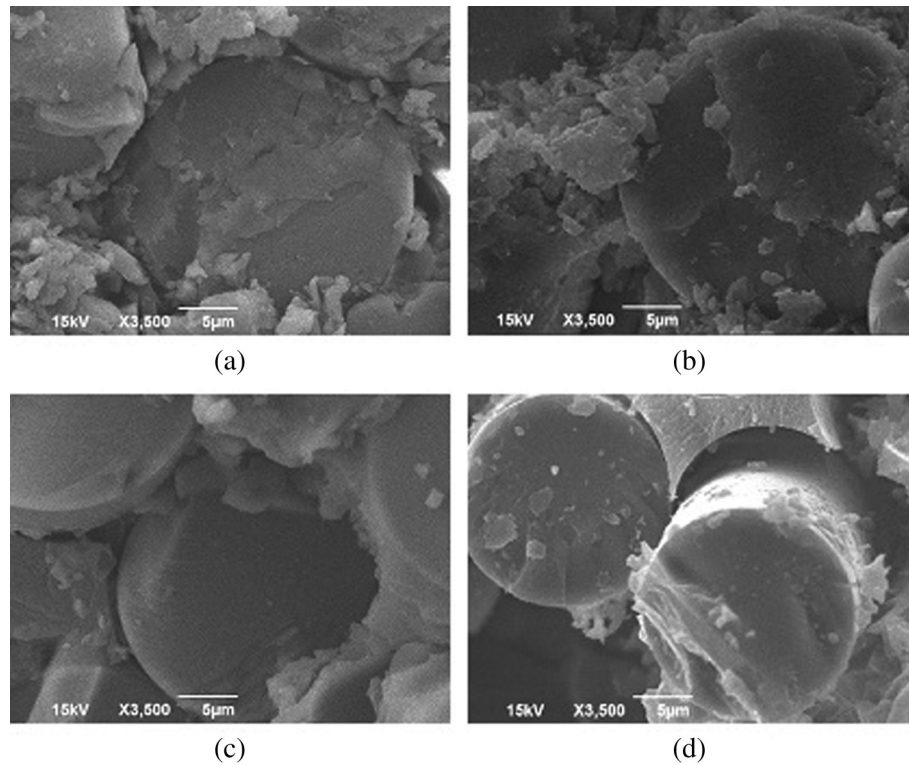


Fig. 12. Cross-sectional micrograph of Type II GFRP bars: (a) control, (b) conditioned for 15 months at 20 °C, (c) conditioned for 15 months at 40 °C, (d) conditioned for 15 months at 60 °C.

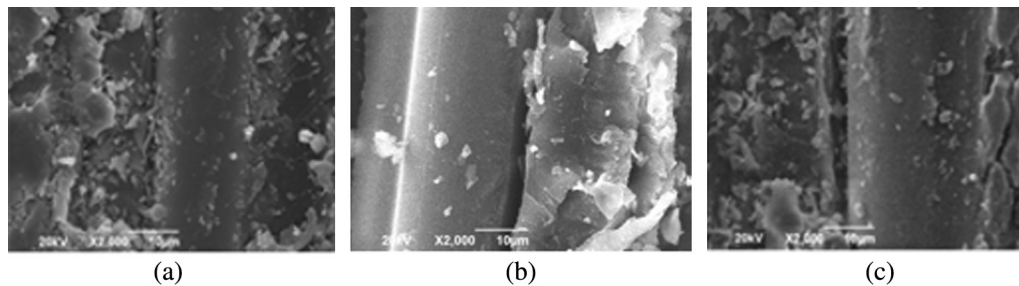


Fig. 13. Longitudinal micrograph of Type I GFRP bars: (a) conditioned for 5 months at 40 °C, (b) conditioned for 10 months at 40 °C, (c) conditioned for 15 months at 40 °C.

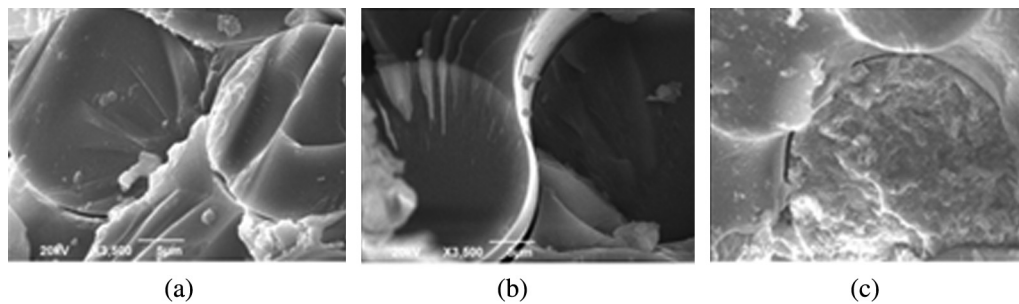


Fig. 14. Cross-sectional micrograph of Type I GFRP bars: (a) conditioned for 5 months at 40 °C, (b) conditioned for 10 months at 40 °C, (c) conditioned for 15 months at 40 °C.

interface due conditioning. Although conditioning at 20 °C initiated a gap at the fiber-matrix interface, the separation between the fiber and matrix was intensified at the higher temperatures. Fig. 12(d) shows that specimens conditioned at 60 °C experienced complete fiber debonding and severe matrix disintegration. Conditioning of GFRP bars at temperatures above 50 °C promotes expansion of

matrix in the transverse direction which could jeopardize the integrity of GFRP bars and impair the bond at the fiber-matrix interface [36–38].

Longitudinal and cross-sectional micrographs of Type I GFRP bars conditioned at 40 °C at different conditioning durations are compared in Figs. 13 and 14, respectively. Similarly, corresponding

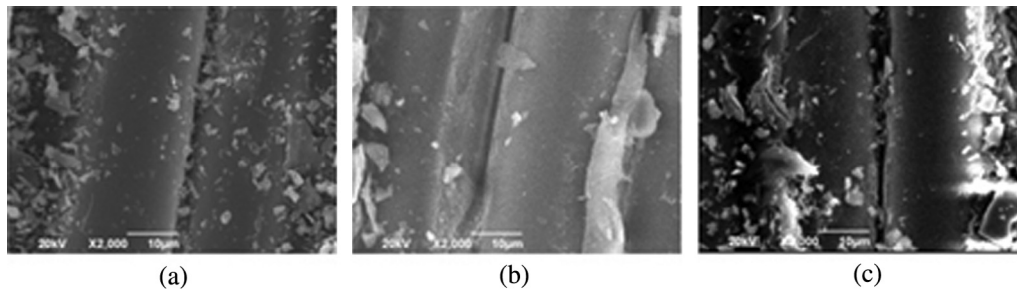


Fig. 15. Longitudinal micrograph of Type II GFRP bars: (a) conditioned for 5 months at 40 °C, (b) conditioned for 10 months at 40 °C, (c) conditioned for 15 months at 40 °C.

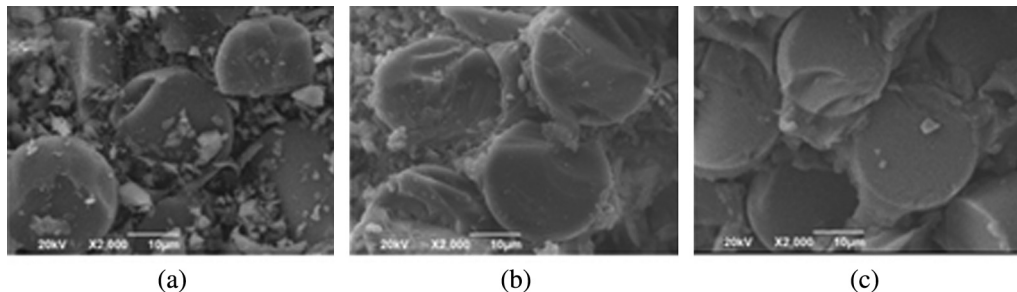


Fig. 16. Cross-sectional micrograph of Type II GFRP bars: (a) conditioned for 5 months at 40 °C, (b) conditioned for 10 months at 40 °C, (c) conditioned for 15 months at 40 °C.

micrographs of Type II GFRP bars exposed to 40 °C over the span of 15 months are shown in Figs. 15 and 16, respectively. Although the specimens experienced some separation between the fiber and matrix due to conditioning, the deterioration at the fiber-matrix interface was not severely intensified with longer conditioning periods. This indicates that the deterioration at the fiber-matrix interface is less sensitive to the conditioning duration rather than the conditioning temperature.

3.4. Matrix digestion analysis

To evaluate the degradation mechanism and assess the conditioning-induced mass dissolution of GFRP bars, matrix digestion by nitric acid was used. Fig. 17(a) and (b) show the respective matrix retention of Type I and Type II GFRP bars after conditioning. Longer conditioning and higher temperatures tended to reduce the matrix retention. This is consistent with other published test results [39]. At 5 months of conditioning, Type II GFRP was more prone to matrix mass loss than Type I, especially at higher temperatures. After 15 months of exposure at 60 °C, Type I GFRP experienced a matrix retention of 83%, while Type II GFRP featured a matrix retention of 62% only. It is possible that the degraded matrix dissolved in the water solution or remained inside the concrete after conditioning. In fact, SEM micrographs of GFRP provide evidence of a relatively clean fiber surface after conditioning for 15 months at 60 °C, indicating lesser polymer matrix at the fiber-matrix interface. A similar phenomenon was reported elsewhere with samples exhibiting a weight loss of 18% after a 2-h exposure at 350 °C [5]. TGA was also employed, according to ASTM E1868 [25], to validate the results obtained by matrix digestion. Table 3 summarizes the matrix content (% by mass) measured using both methods for control samples and those conditioned at 60 °C for 15 months. Matrix retention results obtained by both test methods were similar. The matrix retention after 15 months of conditioning at 60 °C was approximately 83–86% for Type I GFRP bars and 62–64% for Type II bars. Obviously, Type II GFRP bars were more susceptible to degradation during conditioning as manifested by their lower matrix retention.

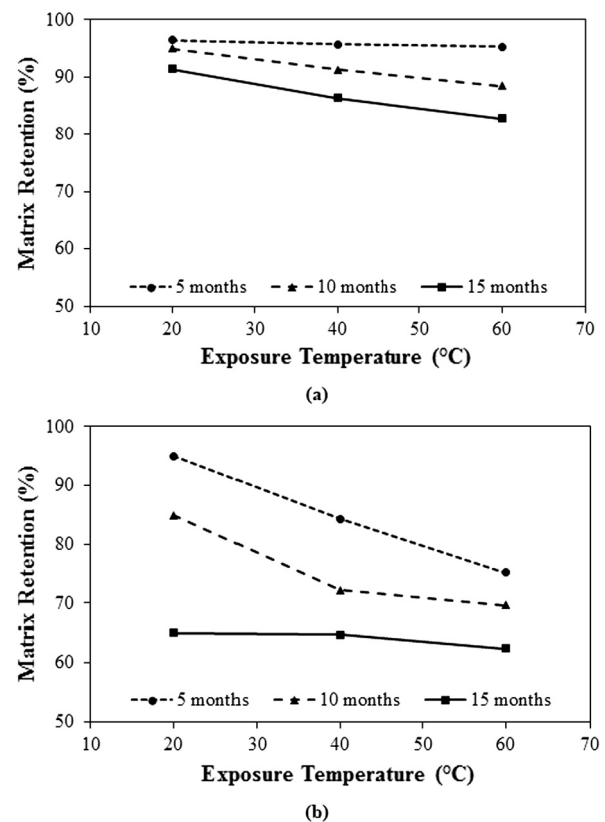


Fig. 17. Matrix retention of GFRP bars: (a) Type I, (b) Type II.

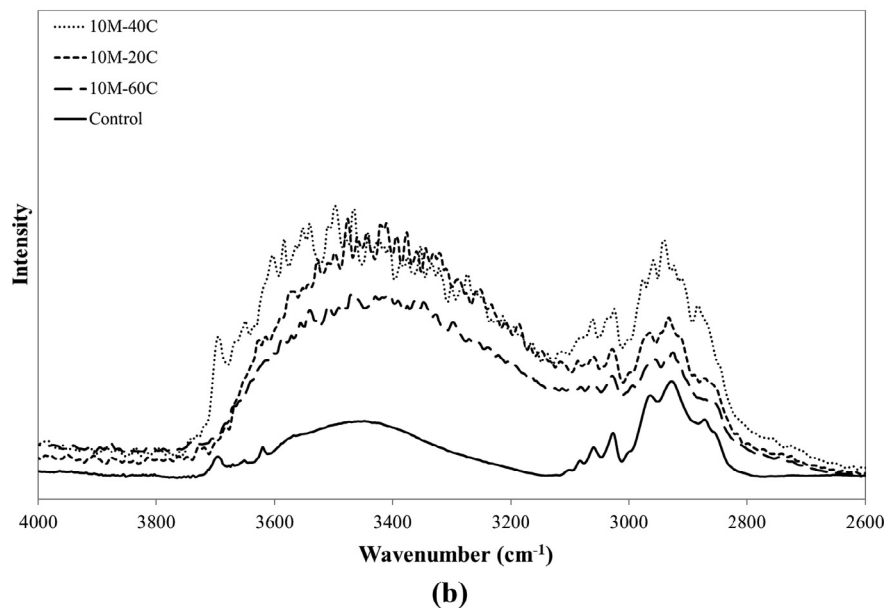
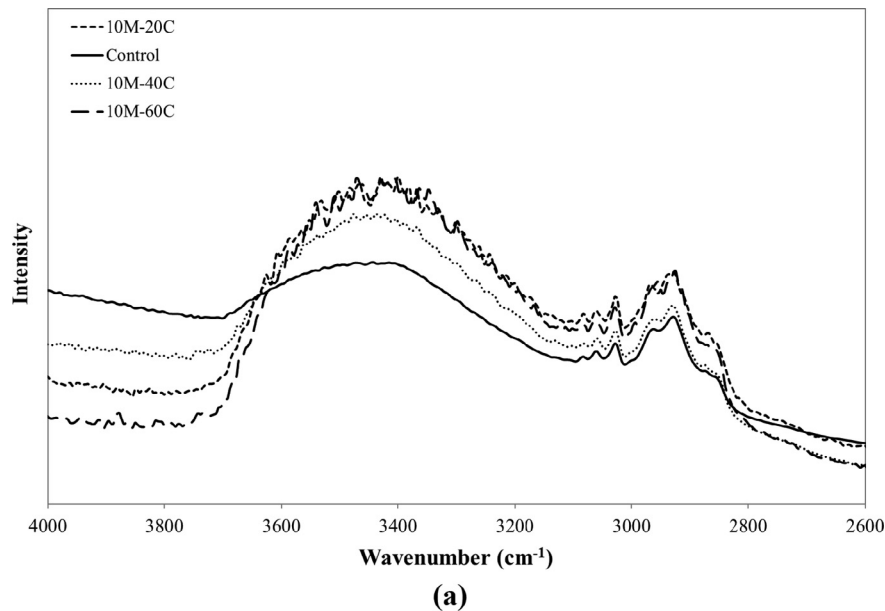
3.5. FTIR analysis

FTIR spectra of Type I and Type II control samples along with those of specimens conditioned for 10 months at different temperatures are plotted in Fig. 18(a) and (b), respectively. FTIR was used

Table 3

Fiber and matrix content of Type I and II GFRP bars using matrix digestion and TGA.

GFRP type	Conditioning		Content by matrix digestion (%)		Content by TGA (%)		Matrix retention (%)	
	Duration (month)	Temp. (°C)	Fiber	Matrix	Fiber	Matrix	Nitric acid	TGA
Type I	Control	Ambient	78.3	21.7	79.0	21.0	–	–
	15	60	82.0	18.0	82.0	18.0	83	86
Type II	Control	Ambient	75.5	24.5	75.0	25.0	–	–
	15	60	84.7	15.3	84.0	16.0	62	64

**Fig. 18.** FTIR spectra of 10-month conditioned GFRP bars: (a) Type I, (b) Type II.

to evaluate the degree of hydrolysis reaction. In the process, two regions were studied: $2800\text{--}3000\text{ cm}^{-1}$, representing carbon-hydrogen groups, and $3200\text{--}3600\text{ cm}^{-1}$, representing hydroxyl groups. While hydrolysis leads to an increase in the infrared band of OH, it does not affect that of CH. The ratio of maximum peaks in each of OH and CH band characterizes the relative amount of hydroxyl groups in the specimen [11,13,40,41]. The OH-to-CH ratio

(OH/CH) of conditioned samples are presented in Table 4. Increasing the conditioning temperature and/or duration led to an increase in the OH/CH, signifying the progression of hydrolysis reaction. For instance, the increase in temperature from 40 to 60 °C for Type I GFRP conditioned for 10 months resulted in an increase in OH/CH from 1.14 to 1.24. Alternatively, when the time of conditioning was extended from 10 to 15 months, Type I GFRP

Table 4
Band ratios of conditioned and control samples.

GFRP type	Conditioning		Peaks		OH/CH	Increase (%) [*]	
	Duration (months)	Temp. (°C)	OH	CH			
Type I	Control	Ambient	2.38	2.23	1.07	–	
	5	20	2.03	1.86	1.09	2	
		40	2.52	2.26	1.12	5	
		60	3.21	2.36	1.36	27	
		10	20	2.62	2.36	1.11	4
	10	40	2.43	2.13	1.14	7	
		60	5.11	4.13	1.24	16	
		15	20	2.26	2.02	1.12	5
			40	4.36	3.75	1.16	8
	60		5.24	4.23	1.24	16	
	Type II	Control	Ambient	0.13	0.21	0.59	–
		5	20	2.86	2.61	1.10	86
40			2.56	2.29	1.12	90	
60			2.46	2.16	1.14	93	
10			20	1.57	1.45	1.08	83
		40	3.74	3.33	1.12	90	
		60	2.82	2.46	1.15	95	
15		20	2.02	1.82	1.11	88	
		40	3.16	2.72	1.16	97	
		60	1.55	1.35	1.15	95	

^{*} Percent increase in the OH/CH ratio of conditioned samples with respect to that of the corresponding control sample.

samples exposed to 40 °C experienced an increase in ratio from 1.14 to 1.16. This shows that increasing the conditioning temperature could further enhance the reaction. It is conclusive that temperature is the more governing parameter compared to time of conditioning. This is consistent with similar observations reported by other researchers [6,13].

The percent increase in the OH/CH ratio of the conditioned GFRP samples relative to that of the corresponding control unconditioned samples is presented in Table 4. In comparison to Type I GFRP bars, Type II counterparts developed much more conditioning-induced hydroxyl groups. As a result, the tensile strength retention was much lower in Type II GFRP. It should be noted that the major increase in OH band for Type II samples was at 5 months. No or insignificant additional increase in the band was recorded over the subsequent 10 months.

3.6. DSC analysis

The glass transition temperature (T_g) of control and conditioned Type I and II GFRP samples are presented in Table 5. Two scans were conducted for each specimen. Samples subjected to higher conditioning temperatures recorded lower T_g values after the first scan. This could be due to matrix plasticization. Evidenced by SEM imaging, higher temperatures intensified the separation between the fiber and matrix and resulted in more voids in the matrix microstructure. Such voids increased the exposure of the polymer matrix to water/moisture and increased the water absorption. As a result, the polymer structure was modified and the T_g was reduced. Additionally, extending the conditioning time from 5 to 10 months while maintaining the same temperature resulted in a lower T_g after the first scan, particularly for Type II GFRP bars. T_g values recorded after 15 months of conditioning, however, were almost equal to those recorded after 10 months, indicating that water absorption had reached an equilibrium state after 10 months of conditioning.

As the samples were heated during the second scan, water evaporation reversed the plasticizing effect, and hence, T_g values of conditioned samples recorded in the second scan became equal to those of the control ones. It should be noted that, for the Type I GFRP control specimen, the T_g corresponding to the second scan was slightly higher than that of the first scan. This could be due to a post-curing phenomenon during the second heating scan. Sim-

Table 5
Glass transition temperature of GFRP bars using DSC analysis.

GFRP type	Conditioning		T _g (°C)		
	Duration (months)	Temperature (°C)	1st scan	2nd scan	
Type I	Control	Ambient	101	106	
		5	20	98	107
			40	98	106
			60	96	107
	10		20	98	107
			40	97	106
			60	95	106
	15		20	98	105
			40	98	106
			60	95	105
	Type II	Control	Ambient	125	125
			5	20	115
			40	105	126
			60	98	126
10			20	100	125
			40	94	126
			60	90	126
15			20	99	126
			40	94	128
			60	90	126

ilar observations were reported by other researchers [4]. Since the T_g after the second scan remained unaffected by conditioning, the polymer matrix did not undergo irreversible chemical degradation. It can be concluded that the deterioration mechanisms of the GFRP bars tested in this study are degradation at the fiber-matrix interface, plasticization, hydrolysis and loss of the polymer matrix during conditioning.

4. Conclusions

The durability performance of two types of commercially-produced GFRP bars conditioned in moist seawater-contaminated concrete has been examined in this paper through microstructural characterization and measurements of tensile strength of conditioned and unconditioned specimens. All specimens failed by rupture of fibers accompanied by fiber debonding at the fiber-matrix interface within the test region. Based on the test results, the following conclusions are drawn:

1. The aging-related degradation in the tensile strength of GFRP bars in moist seawater-contaminated concrete was dependent on the void content and moisture absorption properties. Type II GFRP bars had higher void content and moisture uptake than those of Type I. The increased moisture absorption facilitated progression of the hydrolysis reaction, reduced matrix retention, impaired the bond at the fiber-matrix interface, and hence, Type II exhibited inferior durability performance than that of Type I.
2. Superior short-term tensile properties were not indicative of better durability characteristics. Type II GFRP bars, with the higher initial tensile strength, showed inferior tensile strength retentions than those of Type I, with the lower initial tensile strength. The tensile strength reduction caused by accelerated aging was in the range of 2–15% for the GFRP bar Type I, and 19–50% for GFRP bar Type II.
3. The degradation in properties of GFRP bars was more sensitive to the conditioning temperature rather than conditioning duration. Increasing the conditioning temperature reduced the tensile strength retention at all times of conditioning for both types of GFRP. Increasing the conditioning duration had an almost no effect on the tensile strength retention of Type I GFRP bars, except at the higher temperature of 60 °C where a minor additional strength reduction of 5% was recorded at 15 months of exposure. For Type II, increasing the conditioning time from 5 to 10 months reduced the tensile strength retention. Further increase in the conditioning duration had no or insignificant effect on the tensile strength retention of Type II.
4. Interfacial separation, matrix disintegration, and microgaps were detected in both types of GFRP bars due to conditioning. The matrix disintegration and fiber debonding were intensified with an increase in the conditioning temperature. Type II GFRP bars were more prone to matrix mass loss than Type I, especially at the higher temperatures. After 15 months of conditioning at 60 °C, Type I GFRP experienced a matrix retention of 83%, while Type II GFRP featured a matrix retention of 62% only.
5. Absorbed water reduced the T_g corresponding to the first scan due to a plasticizing effect. Plasticization, however, was more affected by temperature than time of conditioning. The T_g of conditioned specimens after the second scan was almost equal to that of the control specimens. This provided an evidence to the absence of irreversible chemical degradation.

Results of the present study provided insight into the durability performance of two different types of commercially-produced GFRP bars in moist seawater-contaminated concrete. Aging-related degradation of GFRP composite bars in moist seawater-contaminated concrete is highly dependent on the void content and moisture absorption properties which are affected by manufacturing procedure, chemical composition of the matrix, characteristics of the interface, and interfacial imperfections that may develop during the manufacturing process. Tensile strength results reported in the present study are consistent with those of the moisture uptake, matrix digestion, and FTIR tests. The agreement between tensile strength results and those of other microstructural tests conducted on both types of GFRP confirms the credibility and validity of the tensile strength test results. Future research will focus on studying the durability of both types of GFRP bars under a sustained load.

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